Week 2 Day 4 Compose App Lab

Purpose

이 실습은 Day 3에서 길게 입력한 `docker run` 옵션을 `compose.yaml` 로 옮긴다. 핵심은 파일 하나로 web, db, network, volume, environment, healthcheck, cleanup을 재현하는 것이다.

Setup

```
cd week2/day4/labs/compose-app
cp .env.example .env
```

`.env` 의 `POSTGRES_PASSWORD` 는 로컬 실습용 값으로 바꾼다. 실제 운영 secret이나 개인 password를 쓰지 않는다.

Run

```
docker compose config
docker compose up -d
docker compose ps
curl -I http://localhost:18084
curl -s http://localhost:18084 | grep compose-site-v1
docker compose logs db
docker compose run --rm db-client
```

Expected Evidence

```
web-1   running
db-1    healthy
HTTP/1.1 200 OK
compose-site-v1
current_database | current_user
paperclip        | paperclip
```

Failure Drill

`.env` 에서 `POSTGRES_PASSWORD` 줄을 제거한 뒤 config를 확인한다.

```
docker compose config
```

기대 결과는 `set POSTGRES_PASSWORD in .env` 메시지가 포함된 실패다.

Cleanup

```
docker compose down
docker compose down -v
```

`docker compose down` 은 container와 network를 정리한다. `docker compose down -v` 는 named volume까지 삭제하므로 DB data가 사라진다.

File Map

Path	Purpose
`compose.yaml`	web, db, db-client service 정의
`.env.example`	local `.env` 작성 기준
`html/index.html`	nginx가 제공하는 정적 페이지
`db/init/001_schema.sql`	PostgreSQL 최초 초기화 SQL
`README.md`	실행, 확인, 실패, 정리 기준

Services

Service	Image	Role	Exposed To Host
`web`	`nginx:1.27-alpine`	static HTML server	yes, `${WEB_PORT:-18084}:80`
`db`	`postgres:16-alpine`	PostgreSQL database	no
`db-client`	`postgres:16-alpine`	query/check tool	no, profile tool

`db-client` 는 기본 application service가 아니라 확인용 service다. 필요할 때 `docker compose run --rm db-client` 로 실행한다.

Environment Variables

Variable	Required	Default	Notes
`WEB_PORT`	no	`18084`	host에서 web에 접근할 port
`POSTGRES_DB`	no	`paperclip`	practice database name
`POSTGRES_USER`	no	`paperclip`	practice user
`POSTGRES_PASSWORD`	yes	none	`.env`에서 로컬 값으로 설정

`POSTGRES_PASSWORD` 는 required variable이다. 값이 없으면 `docker compose config` 가 실패한다. 의도적으로 실행 전 실패하게 만들어 secret 누락을 빨리 발견한다.

Recommended Local Setup

```
cp .env.example .env
```

그 다음 `.env` 를 열어 `POSTGRES_PASSWORD` 값을 로컬 실습용 값으로 바꾼다. 실제 개인 password, Docker Hub token, cloud access key를 쓰지 않는다.

Preflight

```
docker version
docker compose version
docker compose config
```

Expected:

```
services:
  db:
```

```
  web:
networks:
  app-net:
volumes:
  pgdata:
```

Detailed Runbook

1. Start

```
docker compose up -d
```

2. Check service status

```
docker compose ps
```

Expected signals:

```
web  Up
db   Up (healthy)
0.0.0.0:18084->80/tcp
```

3. Check web

```
curl -I http://localhost:18084
curl -s http://localhost:18084 | grep compose-site-v1
```

Expected:

```
HTTP/1.1 200 OK
compose-site-v1
```

4. Check DB

```
docker compose logs db
docker compose run --rm db-client
```

Expected:

```
database system is ready to accept connections
current_database | current_user
paperclip        | paperclip
```

Network Model

The host reaches the web service through published port `localhost:18084`.

Containers inside the Compose network reach PostgreSQL through service name `db`.

From	To	Address

host browser/curl	web	`localhost:18084`
`db-client` container	db	`db:5432`
host terminal	db	not published by default

Do not document the DB container IP as the connection contract. Container IPs can change. Use service name `db`.

Volume Model

`pgdata` stores PostgreSQL data at `/var/lib/postgresql/data`.

Command	Result
`docker compose down`	removes containers and network
`docker compose down -v`	also removes named volumes

Use `down -v` only when resetting practice data. Do not use it as the default cleanup for data that must be preserved.

Failure Drill 1: Missing Env

Remove `POSTGRES_PASSWORD` from `.env`.

```
docker compose config
```

Expected failure:

```
set POSTGRES_PASSWORD in .env
```

Fix:

```
cp .env.example .env
```

Then set a local practice password again.

Failure Drill 2: Wrong Port

```
curl -I http://localhost:80
curl -I http://localhost:18084
docker compose ps
```

Interpretation:

- `localhost:80` may fail because the published host port is `18084`.
- `localhost:18084` should return HTTP 200 when web is running.
- This is a host port issue, not an nginx container port issue.

Failure Drill 3: Stale Volume

If you edit `db/init/001_schema.sql` after the database has already initialized, the change may not appear. PostgreSQL init scripts run only during initial database creation.

Practice reset:

```
docker compose down -v
docker compose up -d
```

Warning: this deletes practice DB data.

Short RCA Template

```markdown
## Compose App RCA
- Symptom:
- Failed command:
- Error excerpt:
- Category: config / port / env / network / volume / readiness
- Hypothesis:
- Fix:
- Recheck:
- Prevention:
```

Security Notes

- Do not commit `.env`.
- Do not put real passwords or tokens in `.env.example`.
- Do not paste Docker Hub tokens, MFA codes, cloud access keys, or personal credentials into terminal screenshots.
- Prefer placeholder values in documentation.
- Keep DB port unpublished unless the exercise specifically requires host DB access.

Handoff Section For Student README

````markdown
## Docker Compose
### Setup
```bash
cp .env.example .env
```

### Run
```bash
docker compose config
docker compose up -d
```

### Check
```bash
docker compose ps
curl -I http://localhost:18084
curl -s http://localhost:18084 | grep compose-site-v1
docker compose run --rm db-client
```

### Cleanup
```bash
docker compose down
```

Reset practice DB data:
```bash
````
````

````
docker compose down -v
```
````

Evidence Checklist

- [ ] `docker compose config` succeeded.
- [ ] `docker compose ps` shows web running.
- [ ] `docker compose ps` shows db running or healthy.
- [ ] `curl -I http://localhost:18084` returns HTTP 200.
- [ ] body contains `compose-site-v1`.
- [ ] `db-client` query returns `paperclip`.
- [ ] missing env failure is documented.
- [ ] cleanup command is documented.
- [ ] `down -v` data deletion warning is documented.

Scoring

Item	0	1	2
Config	not run	run only	output interpreted
Web	no check	status only	HTTP and body marker
DB	no check	healthy only	query evidence
Network	no explanation	network name only	service name `db` explained
Volume	no note	volume listed	data lifecycle explained
Security	secret exposed	warning only	`.env.example` and no secret exposure
Handoff	none	commands only	expected results and cleanup warning

Completion Statement

```
This Compose app is complete when another student can clone the files, create `.env`, run `docker
compose up -d`, verify web and DB evidence, and clean up without accidentally deleting data they
intended to keep.
```
```